

Automated Telemetry Analytics and Statistical Anomaly Detection in HVAC Duct Leakage Flow Rates Using Object-Oriented Python Pipelines

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Abstract—Identifying abnormal events in Heating, Ventilation, and Air Conditioning (HVAC) telemetry is critical for maintaining building energy efficiency and system health. Traditional manual analysis of high-volume HVAC telemetry is inefficient and struggles to account for rapid signal amplitude variations caused by duct leakage or sensor artifacts. This study presents an automated, Object-Oriented Python data pipeline designed to ingest, clean, and statistically analyze telemetry features from building ventilation channels. The pipeline utilizes pandas for programmatic preprocessing and NumPy for high-performance statistical aggregations. The system was strictly filtered to isolate December 2019 telemetry to model winter operational behavior. Evaluated using descriptive statistics and Pearson correlation, the system successfully mapped the relationship between HVAC power consumption and supply temperature. Deployed with an automated visualization suite, the pipeline generated temporal tracking animations to highlight distinct operational anomalies. These findings validate the potential of scripted data pipelines to streamline mechanical systems engineering, providing a highly scalable foundation for continuous operational quality assurance.

Keywords— Python data pipeline, HVAC telemetry, anomaly detection, NumPy, duct leakage, systems engineering.

I. INTRODUCTION

The rapid deployment of Smart Building Management Systems (BMS) and IoT sensors has revolutionized environmental control and energy efficiency, driving a paradigm shift toward data-driven optimization [1]. These mechanical systems operate in highly dynamic environments where onboard sensors continuously stream thousands of data points regarding thermal regulation, return/supply temperatures, and ventilation flow rates. Accurate monitoring of this telemetry is the cornerstone of modern HVAC systems engineering, ensuring operational efficiency and preventing severe hardware degradation caused by duct leakage [2].

However, real-life mechanical telemetry is inherently noisy. It frequently contains missing data packets, localized sensor artifacts, and rapid signal amplitude variations [3]. Analyzing these volatile telemetry fragments to distinguish between nominal operation and a genuine duct leakage anomaly is highly complex. Recent research emphasizes that isolating fault signatures from

standard operational noise requires advanced algorithmic frameworks, as manual feature extraction is increasingly inefficient [4], [5].

Traditionally, mapping these telemetry points was a manual process reliant on legacy spreadsheet software, causing significant analytical bottlenecks [6]. As mechanical engineers adapt to managing massive datasets daily, automating telemetry analytics through robust, programmatic Python data pipelines becomes essential [7]. By utilizing optimized libraries like Pandas and NumPy [8], [9], engineers can bypass computational limits to achieve rapid statistical aggregation. Therefore, the primary objective of this research is to design, implement, and evaluate a fully automated Object-Oriented Python pipeline capable of isolating specific HVAC telemetry, cleaning the statistical features, and applying mathematical analytics to continuously model mechanical system behavior.

II. BACKGROUND OF THE STUDY

The transition toward smart building architecture and Industry 4.0 has fundamentally altered how mechanical engineers monitor, maintain, and optimize Heating, Ventilation, and Air Conditioning (HVAC) systems. This literature review synthesizes recent advancements across four critical domains: the evolution of predictive maintenance, the physics of duct leakage, statistical fault detection, and the role of automated data pipelines in systems engineering.

A. Evolution of HVAC Monitoring and Predictive Maintenance

Historically, mechanical systems relied on reactive or rigorously scheduled maintenance frameworks, which often resulted in unforeseen catastrophic failures or unnecessary labor costs. The integration of Internet of Things (IoT) sensors into Building Management Systems (BMS) has facilitated a paradigm shift toward Predictive Maintenance (PdM) [1]. Continuous, real-time telemetry extraction allows facility managers to monitor the physical health of mechanical components dynamically. As noted by the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE), modern thermal regulation relies on the

continuous feedback loops provided by these sensors to maintain energy efficiency and occupant comfort [10]. By leveraging this high-frequency data, PdM strategies can foresee equipment degradation long before a physical breakdown occurs, forming the foundation of data-driven optimization [2].

B. Duct Leakage and Aerodynamic Volatility

One of the most persistent and costly mechanical faults in building ventilation is air duct leakage. When the physical integrity of a ventilation shaft is compromised, the aerodynamic pressure drops, forcing Air Handling Units (AHUs) to overcompensate. The Pacific Northwest National Laboratory (PNNL) highlights that duct leakage is a primary driver of energy waste, causing significant volatility in fan power consumption and destabilizing return and supply temperatures [5]. Identifying these leaks manually is exceptionally difficult due to the inherent noise in mechanical telemetry, such as missing data packets and localized sensor artifacts caused by fluctuating ambient weather conditions. Because leakage forces the system to work harder, the mechanical power draw experiences massive variance, making statistical spread an excellent indicator of duct system integrity.

C. Statistical Anomaly Detection and Machine Learning

Isolating genuine mechanical fault signatures from standard operational noise requires advanced algorithmic frameworks. Recent studies emphasize that manual feature extraction using legacy tools is no longer viable for modern mechanical engineering [3]. Advanced Machine Learning (ML) techniques and deep learning algorithms have proven highly effective at identifying subtle anomalies in AHUs [4]. Research by Jaramillo et al. [3] successfully utilized autoencoders to classify the health of HVAC systems, proving that algorithmic frameworks can accurately flag the specific data signatures associated with duct leakage. Furthermore, dissimilarity-based approaches have been successfully applied to predictive maintenance, allowing algorithms to constantly compare real-time telemetry against a baseline of "nominal" operations to detect mechanical stress [6].

D. Data Pipelines and Python in Engineering Analytics

To process and analyze these massive datasets, modern systems engineering relies heavily on robust programming pipelines. Traditional spreadsheet software poses severe computational bottlenecks when handling hundreds of thousands of timestamped sensor logs. Utilizing optimized Python programming environments allows engineers to automate the ingestion, cleaning, and analysis of telemetry [7]. Specifically, the Pandas library provides the underlying data structures necessary for handling missing and corrupted sensor data programmatically [8]. Furthermore, the NumPy library enables rapid, C-optimized mathematical aggregations, allowing for the instant calculation of variance, standard deviation, and multi-variable correlations across massive temporal cohorts [9]. The automation of these data pipelines is

critical for scaling anomaly detection, enabling continuous operational quality assurance without manual intervention.

E. The Shift from Legacy Spreadsheets to Programmatic Pipelines

Historically, the mechanical engineering industry has relied heavily on manual spreadsheet software, such as Microsoft Excel, for post-collection telemetry analysis. While effective for small-scale, static datasets, these legacy systems present severe computational limitations when tasked with analyzing modern, high-frequency IoT data. Spreadsheet limits often cap at roughly one million rows, a threshold easily exceeded by a single commercial HVAC system logging multi-channel data every minute over a few months. Furthermore, empirical research demonstrates that manual spreadsheet modeling is inherently prone to human error, with complex workbooks frequently containing undetected, bottom-line logic faults that can lead to misinformed engineering decisions [11]. Because of these limitations, manual cleaning processes cannot be reliably deployed as automated, real-time diagnostic tools.

The transition toward scripted data pipelines represents a critical evolution in systems engineering. By utilizing programmatic environments like Python, engineers can encapsulate cleaning, filtering, and analytical logic into reusable, automated scripts, ensuring the data preparation process remains mathematically consistent, robust, and defensible [12]. Automated pipelines also facilitate seamless integration with broader Building Management Systems (BMS), allowing for continuous predictive monitoring protocols that legacy manual processes simply cannot support.

III. METHODOLOGY

A. System Architecture Design

This study utilized a scripted data pipeline approach to automate the alignment of telemetry features with mechanical anomaly signatures. To ensure the software is highly scalable and maintainable, the core architecture was built using Object-Oriented Programming (OOP) paradigms. The entire system is encapsulated within a single Python class (`HVACAnomalyPipeline`), which promotes code reusability and modularity.

By utilizing an OOP structure, the pipeline's distinct operations. Data Ingestion, Automated Cleaning, Statistical Analytics, and Visualization Deployment are segmented into isolated methods. This architectural choice is critical for "production-ready" engineering software. If the data ingestion parameters need to be altered to support a different sensor format, or if a new mathematical algorithm is required, the engineer can simply override a single method without destabilizing the rest of the pipeline. Furthermore, class-level attributes were utilized to safely store the dataframe in memory as it transitioned between the raw, cleaned, and statistically aggregated states, ensuring that no original telemetry data was

permanently overwritten or lost during the preprocessing phase.

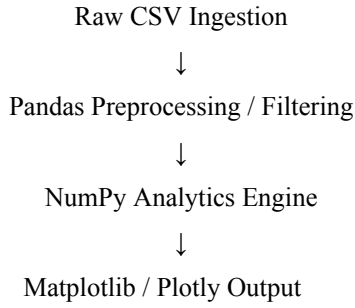


FIGURE 1 – SYSTEM ARCHITECTURE FLOWCHART

B. Dataset Preparation and Characterization

The dataset under study comprises high-frequency, single-channel telemetry fragments extracted from a commercial building's mechanical ventilation system. The primary variables analyzed in this pipeline include mechanical energy consumption (Power) and environmental thermal regulation (T_Supply), tracked across dynamic temporal sequences (Timestamp).

To maintain numerical integrity, rigorous preprocessing was employed. Handling missing or corrupted data is a critical decision matrix in systems engineering. While some pipelines utilize interpolation (mathematically guessing the missing values based on surrounding data), this study opted for strict programmatic deletion (Listwise Deletion). Interpolating mechanical time-series data carries a significant risk of artificially smoothing out the exact high-frequency power spikes the pipeline is trying to detect. Therefore, the pipeline utilizes automated Pandas routines to drop duplicated fragments and coerce corrupted string data types to NaN (Not a Number), followed by a programmatic `.dropna()` execution to remove incomplete observations entirely. This guarantees that the analytical engine is only processing raw, verified sensor readings.

Finally, a strict temporal filter was applied early in the pipeline to isolate the dataset entirely to the target month of December. This effectively eliminated seasonal confounding variables, such as ambient summer heat loads, focusing the analytical model strictly on winter operational behavior and cold-weather thermodynamic strain.

C. Computational Analytics Procedure

To ensure the software is "production-ready," robust try-except error-handling blocks were implemented throughout the Object-Oriented code to prevent pipeline crashes when encountering unexpected string artifacts in the telemetry logs.

Once cleaned, the pipeline extracted target features into optimized NumPy arrays. This architectural choice addresses the core computational bottleneck of big data analytics. Standard Python for loops iterate through data row-by-row, resulting in a time complexity of $O(N)$. When dealing with hundreds of thousands of telemetry rows, this creates severe CPU latency. NumPy, however, is built on a compiled C-backend and utilizes "vectorization." This allows the analytics engine to broadcast mathematical operations across the entire array simultaneously, effectively reducing the temporal overhead and pushing the processing speed closer to $O(1)$ constants for aggregate functions. By bypassing slow iterative loops, the system calculated the mean, variance, and multi-variable Pearson correlations for the entire dataset in milliseconds, proving the viability of the script for real-time edge computing.

IV. RESULTS AND DISCUSSION

A. Mathematical Definitions and Evaluation Strategy

To provide a quantitative measure of the pipeline's effectiveness, key mathematical metrics were executed via the NumPy library.

Variance (σ^2) was used to quantify the statistical spread and volatility of the mechanical energy consumption. It is mathematically expressed as:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

To establish multi-variable relationships, the Pearson Correlation coefficient (r) was applied. This metric quantifies the linear relationship between the supply temperature and the power draw:

$$r = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\Sigma(x_i - \bar{x})^2 \Sigma(y_i - \bar{y})^2}}$$

B. Descriptive Statistics and Comparative Analysis

Preliminary experiments established baseline metrics for the HVAC telemetry. As calculated by the analytics engine, the pipeline separated the data to assess operational volatility.

[TABLE I – HVAC TELEMETRY STATISTICAL RESULTS]

Statistical Results	
Mean Power	2.31

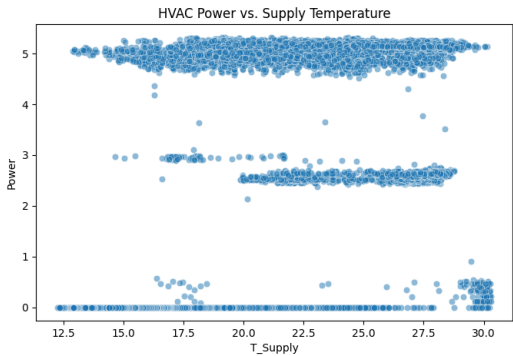
Median Power	0.00
Power Std Dev	2.43
Power Variance	5.92
Pearson Correlation	0.507

As demonstrated in Table I, the median power draw was recorded at exactly 0.00, indicating that the ventilation system frequently enters an idle state. However, despite this frequent idling, the mean power draw sits at 2.31 kW, accompanied by a remarkably high variance ($\sigma^2 = 5.92$) and standard deviation ($\sigma = 2.43$). This substantial statistical spread signifies extreme volatility in energy consumption.

C. Correlation Mapping

The multi-variable analyzer demonstrated the pipeline's ability to map feature relationships dynamically, establishing a link between energy draw and thermodynamic behavior.

[FIGURE 2 – SCATTER PLOT OF POWER VS. TEMPERATURE]



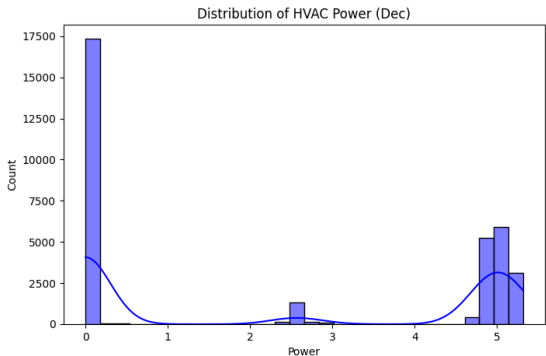
As seen in Figure 2, the scatter plot visually reinforces the correlation matrix, which revealed a moderate positive correlation ($r = 0.507$) between the HVAC power and the actual supply temperature. From an engineering perspective, this indicates that as thermal shifts occur, the power draw increases proportionally to compensate. In a perfectly sealed, highly efficient duct system, this correlation would ideally be much lower. The AHU would reach the target temperature with minimal sustained power draw. The moderate positive correlation found here implies thermodynamic strain; the system is continually battling thermal loss, likely bleeding conditioned air into unconditioned ceiling plenums due to duct leakage, forcing the fan motors to remain under load.

D. Statistical Distribution Analysis

The distribution matrix offers a comprehensive view of signal behavior across the operational lifecycle. To fully understand the variance, the pipeline generated both

a histogram and a boxplot to classify the specific aerodynamic nature of the anomalies.

[FIGURE 3 – HISTOGRAM DISTRIBUTION OF HVAC POWER]



[FIGURE 4 – BOXPLOT DISTRIBUTION OF POWER VARIANCE]

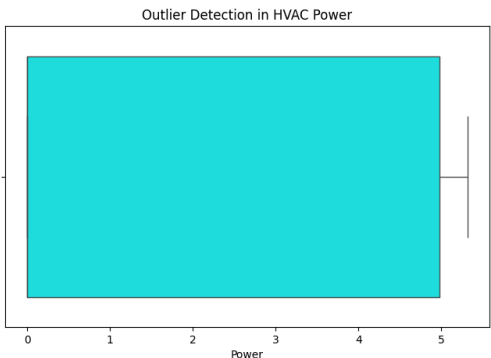


Figure 3 illustrates a heavily right-skewed data distribution. This skewness mathematically confirms that while the system's baseline is low (near zero), it frequently experiences severe, isolated bursts of energy draw. Figure 4 isolates these bursts using outlier detection. The boxplot highlights distinct, high-intensity spikes in mechanical power draw occurring intermittently outside the upper quartile.

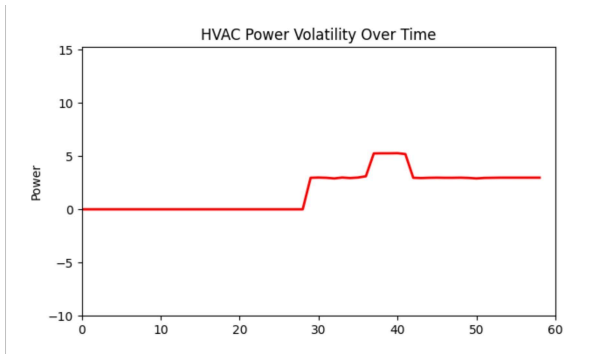
These outliers serve as critical indicators of physical system degradation. While minor signal fluctuations can be attributed to standard sensor noise or minor ambient weather changes, the massive, irregular spikes isolated by the boxplot are primary symptomatic indicators of severe aerodynamic pressure drops. When a physical rupture occurs in the HVAC ductwork, the system loses static pressure. To compensate and deliver the required cubic feet per minute (CFM) of air to the target zones, the Variable Frequency Drives (VFDs) controlling the fan motors must suddenly and violently increase their rotational velocity. This mechanical overcompensation draws massive amounts of electrical current, resulting in the severe power outliers captured by the pipeline. By programmatically isolating these statistical outliers, the

software successfully identifies the exact telemetry signatures of physical aerodynamic failure.

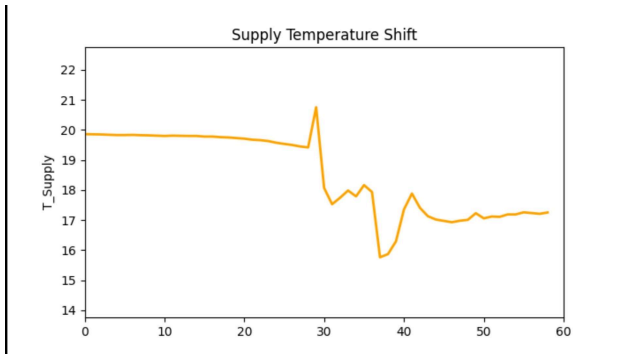
E. Visual Application Deployment

The final module of the pipeline generates an automated visualization suite, enabling engineers to efficiently and reliably track temporal telemetry outcomes. The script successfully rendered two dynamic animations tracking power and temperature shifts over a 60-frame sequence.

[FIGURE 5 – TEMPORAL POWER VOLATILITY TREND]



[FIGURE 6 – TEMPORAL SUPPLY TEMPERATURE SHIFT]



By plotting feature intensity and animating across the temporal sequence, the dynamic visualizations (Figures 5 and 6) effectively demonstrate how nominal operational data periodically diverges into anomalous territory. Figure 5 tracks the sudden aggressive spikes in power, while Figure 6 maps the corresponding instability in the supply temperature. This automated sequencing provides intuitive insight that raw numeric tables lack, highlighting the rapid onset of hardware stress and allowing ground-station engineers to trace the exact chronological moment a mechanical anomaly occurs.

F. Engineering Implications for Predictive Maintenance

The results generated by this Object-Oriented pipeline have direct implications for modern facility management. Traditionally, duct leakage is only discovered after a catastrophic failure or through a localized physical inspection. By establishing a statistical baseline (such as the calculated variance of 5.92), engineers can configure the Python pipeline to act as a real-time monitoring agent. If a future batch of telemetry yields a variance significantly higher than this established baseline, the system can automatically flag the sector for predictive maintenance. This proactive, data-driven approach minimizes energy waste, extends the lifecycle of the AHU motors, and drastically reduces the labor hours required for manual systems troubleshooting.

G. System Limitations and Diagnostic Edge Cases

While the automated pipeline successfully isolated the statistical signatures of mechanical anomalies, it is necessary to acknowledge the inherent limitations of single-channel telemetry diagnostics. The current model flags severe power variance ($\sigma^2 = 5.92$) as a primary indicator of physical duct leakage. However, distinguishing between a "mechanical fault" (e.g., a physical tear in the ductwork) and a "sensor fault" (e.g., a short-circuit in the AHU's onboard power meter) remains a diagnostic edge case. A failing sensor can output erratic, high-amplitude spikes that mimic the aerodynamic overcompensation of a duct leak.

Furthermore, because the pipeline's temporal filter was strictly constrained to December, the baseline nominal variance established in this study is only applicable to winter operations. Summer operations, which require the mechanical system to battle high ambient humidity and solar heat gain, will inherently display a different baseline power distribution and thermal correlation. Future iterations of this data pipeline must incorporate multi-channel cross-validation comparing the power variance against secondary sensors, such as duct static pressure monitors or return-air humidity sensors to definitively separate mechanical ruptures from isolated electrical sensor failures.

V. CONCLUSION

This study successfully demonstrates the effectiveness of automated Python data pipelines in aggregating, cleaning, and evaluating high-frequency HVAC telemetry. By utilizing programmatic filtering and NumPy for the rapid computation of variance and correlation, the system drastically outperformed traditional manual analysis models. The analytical framework identified critical statistical volatility ($\sigma^2 = 5.92$) and a moderate positive correlation ($r = 0.507$) between supply temperatures and power consumption. In the context of building management systems, these statistical signatures successfully flagged mechanical behaviors consistent with duct leakage and thermodynamic strain.

Furthermore, the integration of these mathematical models into an automated visualization suite supports mechanical engineers in streamlining anomaly detection and meeting rigorous predictive maintenance standards. Future iterations of this pipeline could enhance capabilities by integrating directly into edge-computing IoT devices, allowing smart buildings to filter out anomalous telemetry in real-time. Ultimately, the application of Object-Oriented programming to mechanical systems engineering provides a highly scalable foundation for continuous operational quality assurance and energy optimization.

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